YAKEEN

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Physics | Lecture - 9

Basic Maths and Calculus

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Todays Goal

PhD on differentiation

$\left\{ \frac{d}{dx} \right\}$ = differential operator

$\frac{d\square}{dx}$ = The rate of change in $\square$ w.r.t "x"

$\frac{d\boxed{y}}{dx}$ = The rate of change in $\boxed{y}$ w.r.t x

$\frac{d\boxed{v}}{dx}$ = The rate of change in $\boxed{v}$ w.r.t x

$\left[\frac{d[x]}{dt} \right.$ = The rule of change in "x" w.r.t "t" = velocity

Rule of differentiation:

Rule 1. Differentiation of a constant function.

$$\frac{dy}{dx} = 0 \quad ; \quad y = \underline{\text{constant}}$$

$$\frac{d[\sin\theta \cdot \operatorname{cosec}\theta]}{d\theta} = 0$$

$$\frac{d[\pi]}{dt} = 0 \qquad \frac{d\pi}{dx} = 0$$

$$\frac{d(\sin^2\theta + \cos^2\theta)}{d\theta} = 0$$

$$\frac{d(\tan\theta \cdot \cot\theta)}{d\theta} = 0$$

$$\frac{d\,2^\circ}{dt} = 0$$

$$\frac{d(x^0)}{dx} = 0$$

$$\frac{d(\sin 45^\circ)}{d\theta} = 0$$

Rule - 2 Differentiation of algebric function

$$y = x^n$$

$$\frac{dx^n}{dx} = n x^{n-1}$$

① $\frac{dx^3}{dx} = 3x^{3-1}$

$= 3x^2$

② $\frac{dx^5}{dx} = 5x^{5-1} = 5x^4$

$$y = x^{10}$$

$$\Rightarrow \frac{dx^{10}}{dx} = 10 x^{10-1} = 10x^9$$

$$\Rightarrow \frac{dx^{1/2}}{dx} = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2} x^{-1/2}$$

$$\Rightarrow \frac{dx^{3/5}}{dx} = \frac{3}{5} x^{\frac{3}{5}-1} = \frac{3}{5} x^{-2/5}$$ ✓

Rule-3 Diff^n of exponential / Trigo / log.

$\frac{d \sin x}{dx} = \cos x$

$\frac{d \cos x}{dx} = -\sin x$

$\frac{d \tan(x)}{dx} = \sec^2 x$

$\frac{d \sec x}{dx} = \sec x \cdot \tan x$

$\frac{de^x}{dx} = e^x$

$\frac{d \log_e x}{dx} = \frac{1}{x}$ ✓

$\frac{d \cot x}{dx} = -\operatorname{cosec}^2 x$

Rule - 4 If any Constant multiply with Variable then Constant comes out from differentiation.

$$y = 3x\sin(x)$$

$$\odot \quad \frac{dy}{dx} = \frac{d(3\sin(x))}{dx} = 3\frac{d\sin(x)}{dx}$$

$$\odot \quad y = 4e^x \qquad = 3\cos(x)$$

$$\frac{d4e^x}{dx} = 4\frac{de^x}{dx} = 4ex$$

$$\Rightarrow y = 10\,x^{1/9}$$

$$\frac{dy}{dx} = 10\,\frac{1}{9}x^{\frac{1}{9}-1}$$

$$= \frac{10}{9}x^{-8/9}$$

Rule → 5 addition Rule

Ex- $y = A + B$

$$\frac{dy}{dx} = \frac{d(A+B)}{dx}$$

$$\frac{dy}{dx} = \frac{dA}{dx} + \frac{dB}{dx}$$

Ex $y = 4x^3 + e^x$

$$\frac{dy}{dx} = \frac{d\,4x^3}{dx} + \frac{de^x}{dx}$$

$$= 4\frac{dx^3}{dx} + e^x = 4(3x^2) + e^x$$

Rule-6 Subtraction Rule

$$y = A - B$$

$$\frac{dy}{dx} = \frac{dA}{dx} - \frac{dB}{dx}$$

Ex- $y = \log e^x - \sin x$

$$\frac{dy}{dx} = \frac{d\log e^x}{dx} - \frac{d\sin x}{dx}$$

$$= \frac{1}{x} - \cos x$$

Rule-7

Product Rule

$$y = A \cdot B$$

$$\frac{dy}{dx} = \frac{dA}{dx}(B) + A\frac{dB}{dx}$$

Ex

$$y = x^2 \cdot \sin(x)$$

$$\frac{dy}{dx} = \left(\frac{dx^2}{dx}\right)\sin(x) + x^2\frac{d\sin(x)}{dx}$$

$$= 2x\sin(x) + x^2\cos x$$

Ex

$$y = x^5 \times e^x$$

$$\frac{dy}{dx} = \frac{dx^5}{dx}e^x + x^5\frac{de^x}{dx}$$

$$= 5x^4e^x + x^5(e^x)$$ Ans

$$y = \tan x \cdot \log_e x$$

$$\frac{dy}{dx} = \frac{d\tan x}{dx} \times \log_e x + \tan x \times \frac{d\log_e x}{dx}$$

$$\frac{dy}{dx} = \sec^2 x \log_e x + \tan x \frac{1}{x}$$

Rule-8 Division Rule.

$$y = \frac{A}{B}$$

$$\frac{dy}{dx} = \frac{B\left(\frac{dA}{dx}\right) - A\left(\frac{dB}{dx}\right)}{B^2}$$

Ex $y = \frac{e^x}{x^5}$ (division)

$$\frac{dy}{dx} = \frac{x^5 \frac{de^x}{dx} - e^x \frac{dx^5}{dx}}{(x^5)^2}$$

$$\frac{dy}{dx} = \frac{x^5(e^x) - e^x 5x^4}{x^{10}}$$

$$y = \frac{\sin x}{\cos x}$$ find $\frac{dy}{dx}$.

$\cos^2 x = [\cos(x)]^2$

Soln Division Rule

$$\frac{dy}{dx} = \frac{\cos(x)\frac{d\sin x}{dx} - \sin(x)\frac{d\cos x}{dx}}{\cos^2 x}$$

$$= \frac{\cos(x)\times\cos x - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

$y = \frac{\sin x}{\cos x} = \tan x$

$\frac{dy}{dx} = \frac{d\tan x}{dx} = \sec^2 x$

$$y = \frac{A}{B}$$

$$\frac{dy}{dx} = \frac{B\frac{dA}{dx} - A\frac{dB}{dx}}{B^2}$$

$$y = \frac{e^x}{x^5}$$

→ (lars video se le 10

$$y = x^2 \cdot \log x$$

$$\frac{dy}{dx} = 2x \log x + x^2\left(\frac{1}{x}\right)$$

$$y = \frac{\log x}{x^2}$$

$$\frac{dy}{dx} = \frac{x^2\left(\frac{1}{x}\right) - \log x\,(2x)}{x^4}$$

(6) If $y = x^2 - 5x + x^3$ find $\left[\frac{dy}{dx}\right]$ at $x = 1$

Soln

zero $y_{(x=1)} = (1)^2 - 5(1) + (1)^2$

$= 1 - 5 + 1 = -3$



$y = x^2 - 5x + x^3$

$\frac{dy}{dx} = \frac{dx^2}{dx} - 5\frac{dx}{dx} + \frac{dx^3}{dx}$

$= 2x - 5 \times 1 + 3x^2$

$\boxed{\frac{dy}{dx} = 2x - 5 + 3x^2}$

$\left|\frac{dy}{dx}\right|_{x=1} = 2 \times 1 - 5 + 3(1)^2 = 2 - 5 + 3$

$= 0$

(#) $y = \cos x + \frac{2}{x^3} = \cos x + 2x^{-3}$ }

$$\frac{dy}{dx} = \frac{d\cos x}{dx} + 2\frac{dx^{-3}}{dx}$$

$$\frac{dy}{dx} = -\sin x + 2(-3x^{-4})$$

$$\frac{dy}{dx} = -\sin x - 6x^{-4}$$

$$\left[\frac{1}{x^2} = x^{-2}\right.$$

$$\left.\frac{1}{x^3} = x^{-3}\right]$$

$$y = \frac{2}{x} + \frac{3}{x^3} + \frac{4}{x^5} - \frac{6}{x^2} + x$$

$$y = 2x^{-1} + 3x^{-3} + 4x^{-5} - 6x^{-2} + x$$

diff^n wrt x

$$\frac{dy}{dx} = 2(-1x^{-2}) + 3(-3x^{-4}) + 4(-5x^{-6}) - 6(-2x^{-3}) + \frac{dx}{dx}$$

$$= -2x^{-2} - 9x^{-4} - 20x^{-6} + 12x^{-3} + 1$$

$$\frac{dx}{dx} = \frac{-2}{x^2} - \frac{9}{x^4} - \frac{20}{x^6} + \frac{12}{x^3} + 1$$

$$\frac{dx^1}{dx} = 1x^{1-1}$$
$$= 1x^0$$
$$= 1$$

Q gf$_{\text{velocity}}$ $V = 3t^2 - 4t + 5$ find $\left(\frac{dv}{dt}\right)$ at $t = 2$:

Soln

$$\frac{dv}{dt} = 3\frac{d(t^2)}{dt} - 4\frac{dt}{dt} + \frac{d5}{dt}$$

$$= 3 \times 2t^1 - 4 \times 1 + 0$$

$$V = 6t - 4$$

$$V_{t=2} = 6 \times 2 - 4$$

$$= 12 - 4$$

$$= 8$$

Q1 $y = 3x^2 - 2x^3$ find x where $\frac{dy}{dx} = 0$

diff^n w.r.t. x

Soln

$$\frac{dy}{dx} = 3(2x) - 2(3x^2)$$

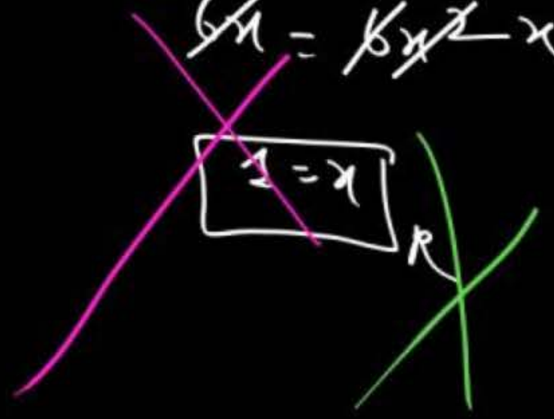
$$\Rightarrow 6x - 6x^2 = 0 \longrightarrow 6x(1-x) = 0$$

$6x = 0$

$x = 0$

$1 - x = 0$

$x = 1$



(Q) $y = \frac{A}{x^3} - \frac{B}{x^4}$ find x where $\left(\frac{dy}{dx} = 0\right)$

$y = Ax^{-3} - Bx^{-4}$

diff^n w.r.t x

$$\frac{dy}{dx} = A(-3x^{-4}) - B(-4x^{-5})$$

$$\frac{dy}{dx} = \frac{-3A}{x^4} + \frac{4B}{x^5} = 0$$

$$\frac{4B}{x^5} = \frac{3A}{x^4}$$

$$x = \frac{4B}{3A}$$

#

$$y = \frac{x^2-1}{x^2+1}$$ diff^n w.r.t. x $\frac{dy}{dx} = ??$

Division Rule

$$\Rightarrow \frac{dy}{dx} = \frac{(x^2+1)\frac{d(x^2-1)}{dx} - (x^2-1)\frac{d(x^2+1)}{dx}}{(x^2+1)^2}$$

$$= \frac{(x^2+1)\times(2x-0) - (x^2-1)[2x+0]}{(x^2+1)^2}$$

$$= \frac{(x^2+1)(2x) - (x^2-1)(2x)}{(x^2+1)^2} = \frac{2x^3+2x - 2x^3+2x}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}$$ ✓

gf $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$

$(a+b)^2 = a^2+b^2+2ab$ ✓

$y = (\sqrt{x})^2 + \left(\frac{1}{\sqrt{x}}\right)^2 + 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}}$

$y = x + \frac{1}{x} + 2$

$y = x + x^{-1} + 2$

↳ diff^n w.r.t x

$\frac{dx}{dx} = 1$ ✗

$\frac{dy}{dx} = 1 - 1x^{-1-1} + 0$

$\frac{dy}{dx} = 1 - \frac{1}{x^2} = \frac{x^2-1}{x^2}$

Rule-9 Product Rule

$$y = ABC$$

diff^n y w.r.t x

$$\frac{dy}{dx} = \left(\frac{dA}{dx}\right)BC + A\left(\frac{dB}{dx}\right)C + AB\left(\frac{dC}{dx}\right)$$

$$y = x^2 \sin x\, e^x$$

$$\frac{dy}{dx} = 2x(\sin x\, e^x) + x^2(\cos x)e^x + x^2 \sin x\,(e^x)$$

$$\frac{de^x}{dx} = e^x$$

Rule-10

Double Differention

$y(x)$

diff^n of y w.r.t x

$\frac{d}{dx}(y)$ = diff^n of y w.r.t x

= slope

= derivative of y w.r.t x

= Rate of change in y w.r.t x

$\frac{dy}{dx}$ → diff^n of $\frac{dy}{dx}$ w.r.t x

$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2}{dx^2}(y)$ = double diff^n y w.r.t x

$\frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3(y)}{dx^3}$ ≠

5++5 = ?? wrong ✗

5+5 = 10 ✓

5+2+5 = 15 ✓

If $y = \sin(x)$

$$\frac{dy}{dx} = \cos(x)$$ = single derivative

↓ Diff^{n} w.r.t x

$$\frac{d^2y}{dx^2} = -\sin(x)$$ → double derivative

$$\frac{d^3y}{dx^3} = -\cos x$$

$$\frac{d^4y}{dx^4} = -[-\sin x) = +\sin x$$

If $y = 4x^3 + 3\sin x - 4e^x$.

find $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$

$$\frac{dy}{dx} = 4[3x^2] + 3[\cos x] - 4(e^x)$$

$$\boxed{\frac{dy}{dx} = 12x^2 + 3\cos(x) - 4e^x}$$

diff^n w.r.t x

$$\frac{d^2y}{dx^2} = 12[2x] + 3(-\sin x) - 4e^x$$

$$= 24x - 3\sin x - 4e^x$$

diff^n w.r.t x

$$\frac{d^3y}{dx^3} = 24 - 3\cos x - 4e^x$$

$$\boxed{\frac{d^4y}{dx^4} = +3\sin x - 4e^x}$$

If $y = 1x^5 + 2x^4 + 3x^3 + 4x^2 + 5x^1 + 6x^0$

diff^n w.r.t x

$\frac{dy}{dx} =$

$\frac{d^2y}{dx^2} =$

$\frac{d^3y}{dx^3} =$

$\frac{d^4y}{dx^4} =$

Chain Rule

$y = \sin(x)\, x^3$ (Product Rule)

$y = \sin(x) + x^2$ ← addit. Rule

outer funct'n ↓ ; inner funct. ↺

$y = \sin(x^3)$ ←

$y = \sin(e^x)$ ←

$y = e^{(\sin x)}$ ←

} function of fun.

Outside - Inside Rule

$\# \quad \frac{dy}{dx} = \left[\frac{\text{diff}^n \text{ of outer function}}{\text{keep inner function as it is}}\right] \times \left[\text{diff}^n \text{ of inner funct}\right]$

$y = \sin(\underbrace{x^3}_{\text{Inner}})$

$\# \quad \frac{dy}{dx} = \cos(x^3) \times \frac{dx^3}{dx}$

$= \cos(x^3) \times 3x^2$

$$y = \sin(\text{PoPu})$$

$$\frac{dy}{dx} = \cos(\text{PoPu}) \times \frac{d(\text{PoPu})}{dx}$$

$$y = e^{(\text{Rambu})}$$

$$\frac{dy}{dx} = e^{(\text{Rambu})} \frac{d(\text{Rambu})}{dx}$$

$$y = (\underset{\uparrow \text{Innr}}{M \cdot R})^2$$

$$\frac{dy}{dx} = 2(MR)^{2-1} \times \frac{d(MR)}{dx}$$

$$x^2 \longrightarrow \frac{dx^2}{dx} = 2(x)^{2-1}$$

$y = \sin(x^3)$

↑ outer ↑ inner

diff^n w.r.t. x

$$\frac{dy}{dx} = \cos(x^3) \times 3x^2$$

$y = \cos(e^x)$

$$\frac{dy}{dx} = -\sin(e^x) \times e^x$$

#

$$y = \sin(2x^3+4)$$

outer — inner

diff^n w.r.t x

$$\frac{dy}{dx} = \cos[2x^3+4] \times \frac{d(2x^3+4)}{dx}$$

$$= \cos(2x^3+4) \times [2(3x^2)+0]$$

$$= \cos(2x^3+4) \times 6x^2$$

$$y = e^{(x^2)}$$

outer: $e^{(\cdot)}$ — inner: x^2

$$\frac{dy}{dx} = e^{(x^2)} \times \frac{d(x^2)}{dx}$$

$$= e^{x^2} [2x]$$

$$= 2x\, e^{(x^2)}$$

$$y = e^{(\sin x)}$$

outer: $e^{(\cdot)}$ — inner: $\sin x$

$$\frac{dy}{dx} = e^{\sin(x)} \times \frac{d(\sin x)}{dx}$$

$$= e^{\sin(x)} \times \cos x$$

$y = (x^2+2x)^3$ ← out-2

Inn-2

out-side Inside

diff^n.

$$\frac{dy}{dx} = 3(x^2+2x)^{3-1} \times \left[\frac{d(x^2+2x)}{dx}\right]$$

$$= 3(x^2+2x)^2 \times (2x+2)$$

$y = (2x+3)^5$

diff^n

$$\frac{dy}{dx} = 5(2x+3)^{5-1} \times (2+0)$$

$$= 5(2x+3)^4 \times 2$$

$$= 10(2x+3)^4$$

If $y = \log_e(\sin x)$

outer: $\log_e$, Inner: $\sin x$

Diff^n of y w.r.t 'x'

$$\frac{d(\log_e x)}{dx} = \frac{1}{x}$$

Soln

$$\frac{dy}{dx} = \frac{1}{\sin x} \times \cos x = \frac{\cos x}{\sin(x)} = \cot(x)$$

(10) $y = \sin(\log_e x)$

diff^n of y w.r.t. 'x'

$$\frac{dy}{dx} = \cos(\log x) \cdot \frac{1}{x}$$

$$y = (x+4)^{1/2} = \sqrt{x+4}$$

Inner

$$\frac{dy}{dx} = \frac{1}{2}(x+4)^{\frac{1}{2}-1} \times \frac{d(x+4)}{dx}$$

$$= \frac{1}{2}(x+4)^{-\frac{1}{2}} \times [1+0]$$

$$= \frac{1}{2}(x+4)^{-1/2}$$

Answer

#

$$y = (\sin x)^2$$ ← outer

↑ inner

Soln

$$\frac{dy}{dx} = 2(\sin x)^{2-1} \times \frac{d\sin(x)}{dx}$$

$$= 2\sin(x)\cos x$$

$$= \sin(2x)$$

#

$$y = \sin(x^2)$$

$$\frac{dy}{dx} = \cos(x^2) \times 2x$$

(e) $y = e^{\sin(x^3)}$

$e^{\sin(x^3)}$ ← outer

$$\frac{dy}{dx} = e^{\sin(x^3)} \times \cos(x^3) \times 3x^2$$

If $y = A\sin(\omega t)$ then find $\frac{dy}{dt}$ & $\frac{d^2y}{dt^2}$??

Inner

Also Prove $\frac{d^2y}{dt^2} + \omega^2 y = 0$

Soln $\frac{dy}{dt} = A\cos(\omega t)\frac{d(\omega t)}{dt}$

$= A\cos(\omega t) \times \omega \times 1$

$\frac{dy}{dt} = A\omega\cos(\omega t)$

$\frac{d^2y}{dt^2} = A\omega\left[-\sin(\omega t)\frac{d(\omega t)}{dt}\right]$

$= -A\omega\sin(\omega t) \times \omega \times 1$

$\frac{d^2y}{dt^2} = -A\omega^2\sin(\omega t)$

Put y

$\frac{d^2y}{dt^2} = -\omega^2 y$

$\frac{d^2y}{dt^2} + \omega^2 y = 0$

eqn of SHM

Partial differential:

$Y(x)$

$y = x^3$

$\frac{dx^3}{dx} = 3x^2$

$V(x, y, z)$ ← V depends of (x, y & z)

Partial diff^n

$$\Rightarrow \left(\frac{dv}{dx}\right)_{y \& z \text{ const}} + \left(\frac{dv}{dy}\right)_{x,z = \text{const}} + \left(\frac{dv}{dz}\right)_{x \& y \text{ const}}$$

$V = x^2 y^3 z^4$ find $\left(\frac{dv}{dx}\right) = ??$

$\left(\frac{dv}{dy}\right)$

$\left(\frac{dv}{dz}\right) = ??$

Par

If $y = 2t^3 + 1$ $\quad x = t^2$ $\quad$ Find $\frac{dy}{dx}$

$y = \sin x \qquad x = 3x \qquad \text{find } \frac{dx}{dy}$

Find slope of equation $y = 3x^2 + 5x - 4$ at $x = 1$

$y = \mathrm{A}\cos(kx)$ $y = 5e^{3x}$ $y = e^{-x}$ $y = 3\sin(x^2 + 2)$

H/W , sirf Ye Question hi Karna hai

Question nos from MR Physics Pdf →

74, 90, 95, 97, 100

102, 104, 105, 106, 115, 116, 117, 119

122, 121, 132, 135, 137, 140

147, 154, 165, 171, 172, 180, 181

182, 184, 185, 186, 188

Thank You